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Two periodization strategies in professional football and their influence on accumulated training load and its distribution: Differences between starters and non-starters

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ABSTRACT

This study compared weekly training-load (TL), training-monotony (TM), and training-strain (TS) in 80 professional football players (28.1 ± 3.2 yrs, 78.0 ± 5.9 kg, and 182.0 ± 4.8 cm), considering two lengths (6 and 7 days between matches), two types of periodization strategies (placing the compensatory session in MD+1 [P1] or MD+2 [P2] post-match-day) and two players' status, starters, and non-starters. Using Global Positioning Technology, the monitored variables were: Player Load (PL), Total Distance (TD), distances at various speeds (>21 [HSR], >24 [VHSR], and > 27 [SPR] km·h^{-1}), number of accelerations (>2 m·s^{-2}, ACC), and decelerations (<-2 m·s^{-2}, DEC). Kruskal-Wallis and Mann-Whitney non-parametric statistical tests were used for loading strategy differences. In both lengths and periodization strategies of the micro-cycles, non-starters trained more than starters. There were differences in the quantity (TL) and distribution (TM and TS) demand considering the micro-cycle's length and periodization strategy. In MD+1 (P1), non-starters accumulated higher TL, showing moderate to large Cohen's d magnitudes in HSR, VHSR, SPR, PL, and TD. For starters, training in MD+2 (P2) resulted in higher values ($p < 0.05$) of TL, TM and TS. The coaching staff needs to anticipate TL, TM, and TS by considering players' status, periodization strategy, and the number of training sessions between matches.

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Football; team sport; monitoring; training demand; GPS

1. Introduction

Analyzing player and team performance in a soccer match is a complex process due to the dynamic non-linear nature of the activity, which emerges from the interaction of multiple levels (e.g., individual, position, and team) and dimensions such as behavioural, cognitive, socio-affective, and energetic (Pons et al., 2019). The conditional demands of soccer matches appear to have increased in the last 15 years (Barrera et al., 2021; Gualtieri et al., 2023) with approximately a 10% increase in the number of accelerations/decelerations, a 30% increase in distance covered above 19.8 km·h^{-1} (HSR), and up to a 50% increase in sprint distance covered (SPR, >25.1 km·h^{-1}). In this highly demanding conditional context, there is a recognized need to address two objectives simultaneously: player recovery in the 48 hours following competition (Cross et al., 2019) and the development of conditional capacity during training days within the competitive micro-cycle (Buchheit et al., 2021). These strategies aim to improve physical performance in competition and promote player adaptation to reduce the risk of injuries related to inadequate adaptation (Malone et al., 2018).

In this regard, optimal in-season micro-cycle planning adopts an inverted U-shaped profile, characterized by a horizontal alternation across three phases: 1) a 48-hour post-match recovery; 2) a 2 to 4-day conditional acquisition phase focusing on strength, endurance, and speed; and 3) a final tapering phase leading up to competition (Buchheit et al., 2018; Castillo et al., 2019; Guridi et al., 2021; Marín & Castellano, 2023). During the acquisition phase, scheduling strength training on MD-4 is common to stimulate ACC/DEC variables, emphasizing mechanical or intensive load (Stevens et al., 2017). On MD-3, endurance prioritizes variables like total distance (TD) and high-speed running (HSR) (Anderson et al., 2016; Cross et al., 2019). According to Buchheit et al (Buchheit et al., 2018) incorporating specific stimuli for sprints and direction changes on MD-2, corresponding to the third day of acquisition, can enhance performance. These stimuli are well-tolerated and have the potential to improve neuromuscular efficiency (Buchheit et al., 2018).

In this context, there is a dynamic and variability in TL influenced by factors such as the type of week, the player's initial status, playing position, training mode, maturation status, and contextual variables related to the match, as well as their differential participation in competition (Teixeira, Branquinho, et al., 2022). For example, Dalen et al. (2016) identified significant differences in speed thresholds, TL and ACC/DEC variables based on players positions during matches. Additionally, Di Salvo et al. (2007) found that midfielders cover more TD than defenders and forwards during matches. Moreover, the player's maturation process appears to have a relative influence on physical performance, particularly in

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high-intensity variables such as SPR and ACC (Teixeira, Alves, et al., 2022). Finally, starters, with more playing time, covered significantly greater distances in HSR and SPR compared to non-starters, according to previous studies (Anderson et al., 2016). However, it is noteworthy that substitutes who entered in the second half recorded increases of 25% in HSR and 63% in SPR during the last 15 minutes compared to the starters (Hills et al., 2018).

While the workload dynamics of the "theoretical starting player" may appear resolved, daily reality highlights one of the major challenges faced by fitness coaches: the variability in TL among team players (Stevens et al., 2017). This has led to the need for a deeper understanding of how competition participation affects player preparation (Anderson et al., 2016; Martín-García et al., 2018; Stevens et al., 2017). In this sense, Baptista et al. (2019) showed that during training, wide defenders achieved 64% of the SPR speed in matches, while central defenders, central midfielders, and central forwards achieved in training 107%, 100%, and 107%, respectively, than in matches. Anderson et al. (2016) described a greater accumulation of TD and, particularly, HSR and SPR in starters compared to non-starters. In line with this, other studies (Martín-García et al., 2018; Stevens et al., 2017) also concluded that non-starters accumulated around one-third less distance in HSR and SPR than starters during sessions at the end of the micro-cycle.

In order to address TL imbalances during the micro-cycle due to varying competition participation, supplementary sessions are commonly implemented for players with no or less than 60 min of participation (Martín-García et al., 2018). Depending on the chosen periodization, the compensatory session can be scheduled on: 1) MD+1, the day after the match, following a structured periodization model (Casamichana et al., 2022; Martín-García et al., 2018); 2) MD+2, or two days after the competition (Clemente et al., 2019) aligned with tactical periodization (Oliveira, 2007). Debates exist on the optimal placement of the rest day and training session after competition, with the first study indicating that scheduling the rest day at MD+2 May be associated with significantly lower non-contact injury rates, up to 2 to 3 times lower (Buchheit et al., 2023).

Therefore, the purpose of this study was to compare the accumulated TL and distribution (TM and TS), differentiating two groups of player status (starters and non-starters) with two post-match planning strategies: P1: training during the MD+1 (and resting MD+2) versus P2: training during the MD+2 (having rested MD+1), and considering two rest days (n-d) turnarounds, 6-d (six days) or 7-d (seven days) between matches. The results emphasize the need to propose personalized intervention strategies, taking into account the conditional requirements based on each player's participation in competition, specifically, but also including others variables like length of the next micro-cycle or the decision about when resting. These findings aim to provide valuable insights for the coaching staff by focusing on the strategic distribution of physical stimuli during the in-season micro-cycles.

2. Methods

2.1. Subjects

Eighty professional outfield football players participated in this study (age: 28.1 ± 3.2 yrs, body mass: 78.0 ± 5.9 kg and stature: 182.0 ± 4.8 cm). Players belonged to a first division squad of the Spanish La Liga club. The players were grouped by the number of minute they played in the previous match: starter ($n = 1607$) more than 60-min played and non-starter ($n = 3104$) less than 60-min played of the match or did not play in the match (Casamichana et al., 2022). This study complied with the Declaration of Helsinki and were approved by the Ethics Committee of the University of the Basque Country (UPV/EHU) (code: M10-2021-153).

2.2. Design

This study examined two types of n-day turnaround weeks: micro-cycles with three training sessions and six days turnaround time 3T-6D) and four training sessions and seven days turnaround time (4T-7D). Two periodization strategies were considered: P1, involving training on MD+1 and rest on MD+2, and P2, resting on MD+1 and training on MD+2. The sessions on the day after the match (MD+1) and two days after (MD+2) led to the players being divided into two training groups (starters and non-starters). The MD+1 session for starters aimed at regeneration with gym work, while non-starters replicated competition loads in field sessions that included small-sided games and specifics tasks to reproduce high-intensity demands (Casamichana et al., 2022). For the MD+2 session, both groups had a common initial part in the field (≈30 min), but non-starters additionally did complementary SSG (≈30 min more). The distribution of sessions based on micro-cycle length and periodization type is described in Table 1.

Table 1. Distribution of training sessions according to the length of the micro-cycle and the type of periodization.

Periodization	MD	MD+1	MD+2	MD-3	MD-2	MD-1	MD	
3T-6D								
P1	match	training	*day-off*	training	training	training	match	
P2	match	*day-off*	training	training	training	training	match	
Periodization	MD	MD +1	MD +2	MD-4	MD-3	MD-2	MD-1	MD
4T-7D								
P1	match	training	*day-off*	training	training	training	training	match
P2	match	*day-off*	training	training	training	training	training	match

3T-6D is three training sessions in six days between matches; 4T-7D is four training sessions in seven days between matches, P1 is periodization type 1, P2 is periodization type 2, MD is match-day, MD+1 is one day after the match, MD+2 is two days after the match, MD-4 is four days before the next match, MD-4 is four days before the next match, MD-3 is three days before the next match, MD-2 is two days before the next match, MD-1 is one day before the next match.

2.3. Physical variables

Training load (TL) was monitored on two dimensions, locomotor and neuromuscular. The players running distances were coded into four categories and three speed thresholds: total distance (TD), and distance accumulated at > 21 (HSR), >24 (VHSR) and > 27 (SPR) all in km·h^{-1}. From a neuromuscular dimension, Player Load (PL, in arbitrary units, au) was computed as the vector magnitude representing the sum of accelerations recorded in the anteroposterior, mediolateral, and vertical planes of movement. In addition, the number of moderate-high intensity accelerations (ACC: >2 m·s^{-2}) and decelerations (DEC: <-2 m·s^{-2}) were also registered.

From the TL, the weekly training monotony (TM), corresponding to the mean of training load during the seven days of the week divided by the standard deviation of the days), and training strain (TS), obtained by the multiplication of weekly accumulated load by the TM, were also calculated for each of the external measures (Foster et al., 2001).

2.4. Methodology

Player activity during each training session was monitored using a portable GPS unit (S5, Catapult Innovations Victoria, Australia) with a sampling frequency of 10 Hz placed inside a mini pocket on a specially designed vest between the shoulder blades. To avoid inter-unit errors, each player used the same device throughout the study (Martín-García et al., 2018). GPS data were extracted using the corresponding software (Openfield 2.4, Canberra, Australia) after each training session. Match-day data were not included in the analysis (number of weeks, 3T-6D = 27 and 4T-7D = 13).

Data were collected from outfield players over four consecutive seasons, excluding players who did not complete a full microcycle. The GPS signal quality was measured using 11.0 ± 0.2 satellites, with a horizontal dilution of precision indicating an accuracy of 0.8 ± 0.2. The average GPS quality was 71.8 ± 4.1%.

2.5. Statistical analysis

For descriptive analysis, the data was presented as means and standard deviations (±SD). Normality and homogeneity of variance was examined by conducting Levene's and Shapiro-Wilk tests, respectively. Due to the data not meeting the assumptions of normality and homoscedasticity, a non-parametric approach was chosen for the analysis. A one-way ANOVA was used (Kruskal-Wallis test) to evaluate differences in dependent variables across various groups, status of the players (starters and non-starters), micro-cycle length (3T-6D and 4T-7D), and type of weekly periodization strategy (P1 or P2) as well as to assess differences between the different status of the players (starters and non-starters) and the different sessions of the week (MD+1, MD+2, MD-4, MD-3, MD-2, and MD-1). In the event of a difference occurring, Mann-Whitney post-hoc tests were used to identify any localised effects. Cohen`s d effect size was used for pairwise comparisons. Thresholds for effect size (ES) statistics were < 0.2, trivial; <0.6, small; <1.2, moderate; <2.0, large; and ≥ 2.0, very large. All data analyses was carried out using *Excel* and the statistical analysis software *JASP version 0.9.2* (University of Amsterdam, https://jasp-stats.org/). The level of significance was set at $p < 0.05$.

3. Results

3.1. Weekly demand according to type of periodization and players status

Table 2 presents the values (mean and ±SD) of TL (TD, PL, HSR, VHSR, SPR, ACC and DEC), training monotony (TM) and training strain (TS) registered from each length of micro-cycle (3T-6D and 4T-7D), two types of weekly post-match periodization (P1 and P2), and distinguishing the players status (starters and non-starters).

Regarding players status, significant differences were observed in most variables (Table 2), with higher values in non-starters compared to starters. The P1 showed higher TL for non-starters but lower for starters in both micro-cycles (3T-6D and 4T-7D). Non-starters in P1 showed differences of moderate to important magnitudes in variables such as HSR, VHSR, SPR, PL, and TD, with respect to P2 in both types of micro-cycles (3T-6D and 4T-7D). TM and TS had similar profile. However, for starters the P2 periodization showed higher values for some TL, TM and TS variables compared to P1, with the magnitude of the differences ranging from trivial to moderate.

3.2. Distribution of training demand according to type of periodization and players status

Figure 1 depicts the distribution of global indicators (PL and TD), locomotor (HSR, VHSR, and SPR), and neuromuscular (ACC and DEC) values obtained from each training session (MD+1, MD+2, MD-3, MD-2, and MD-1) of the 3T-6D in relation to the two groups of players, starters and non-starters.

Figure 2 depicts the distribution of global indicators (PL and TD), locomotor (HSR, VHSR, and SPR), and neuromuscular (ACC and DEC) values obtained from each training session (MD+1, MD+2, MD-4, MD-3, MD-2, and MD-1) of the 4T-7D in relation to the two groups of players, starters and non-starters.

Regarding the distribution of the locomotor and neuromuscular demands of the same training days between the different micro-cycles with the same training session of the week between the different micro-cycles, significant differences were found for post competition session (MD+1 and MD+2) (p <*0.05*). In MD+1, for P1 in 3T-6D and 4T-7D, non-starters in had significantly higher stimulation than starters and. In MD+2, for P2 in 3T-6D and 4T-7D non-starters had significantly higher demand than starters.

Regarding the central days, there were no significant differences observed for MD-4. However, for MD-3, there were significant differences ($p < 0.05$) observed between the starters with strategy P1 and P2. In 3T-6D, the starters with strategy P1 had higher values in PL, TD, HSR, VHSR, and ACC compared to those with strategy P2. Additionally, there was a difference in TD among non-starters between 4T-7D with strategy P1 and 3T-6D with strategy P2.

Finally, for the pre-competitive sessions (MD-2 and MD-1), significant differences ($p < 0.05$) were found. In MD-2, starters using strategy P1 in 4T-7D had higher VHSR and DEC compared

Table 2. Total training loads, training monotony and training strain, mean and standard deviation, regarding the lengths of the micro-cycles (3T-6D and 4T-7D) and type of periodization (P1 and P2), considering players status (starter and non-starters).

			Training Load		Training Monotony		Training Strain	
Micro-cycle	Variables	Periodization	Starter	non-starter	Starter	non-starter	Starter	non-starters
3T-6D	TD (m)	P1	12,485.7 ± 2743.9	17,341.5 ± 3,289.3$^{\$\#♣}$	1.1 ± 0.2	1.6 ± 0.3$^{\$\#♣}$	14,576.4 ± 5,953.0	27,877.2 ± 9,747.9$^{\$\#♣}$
		P2	12,360 ± 2231.1	15,287.9 ± 2,455.3$^{\$\#}$	1.2 ± 0.2	1.4 ± 0.2$^{\$\#}$	22,104.7 ± 6,187.0	15,037.7 ± 4,979.5$^{\$\#}$
	PL (au)	P1	1,288.4 ± 326.7	1,805.6 ± 369.1$^{\$\#♣}$	1.1 ± 0.2	1.5 ± 0.3$^{\$\#♣}$	1,521.5 ± 716.8	2,884.9 ± 995.2$^{\$\#♣}$
		P2	1,305.9 ± 235.7	1,615.4 ± 272.5$^{\$\#}$	1.2 ± 0.2$^{\#}$	1.4 ± 0.2$^{\$\#}$	1,599.7 ± 504.9	2,361 ± 677.5$^{\$\#}$
	HSR (m)	P1	253.9 ± 173.1	527.4 ± 294.9$^{\$\#♣}$	0.5 ± 0.3	0.9 ± 05$^{\$\#♣}$	174.5 ± 184.3	609.4 ± 548.9$^{\$\#♣}$
		P2	328.7 ± 193.1$^{\$}$	417.9 ± 216.0$^{\$\#}$	0.7 ± 0.4$^{\$}$	0.8 ± 0.3$^{\$\#}$	287.8 ± 332.9$^{\#}$	385.2 ± 330.3$^{\$\#}$
	VHSR (m)	P1	79.6 ± 75.2	177.4 ± 139.2$^{\$♣\#}$	0.4 ± 0.3	0.7 ± 0.5$^{\$\#♣}$	10.5 ± 30.4	38.8 ± 78$^{\$\#♣}$
		P2	91.2 ± 83.1	131.6 ± 103.1$^{\$\#}$	0.4 ± 0.3	0.6 ± 0.4$^{\$\#}$	13.3 ± 42.4	21.8 ± 49.6$^{\$\#}$
	SPR (m)	P1	18.1 ± 21.2	46.2 ± 57.7$^{\$♣\#}$	0.2 ± 0.3	0.5 ± 0.5$^{\$\#♣}$	2.0 ± 6.7	9.4 ± 31.3$^{\$\#♣}$
		P2	17.6 ± 24.9	26.1 ± 30.9$^{\$\#}$	0.2 ± 0.3	0.3 ± 04$^{\$\#}$	2.2 ± 8.4	3.6 ± 11.7
	ACC (n)	P1	150.8 ± 42.0	236.6 ± 59.0$^{\$♣\#}$	1.0 ± 0.2	1.4 ± 0.3$^{\$\#♣}$	31.1 ± 38.6	67.3 ± 60.1$^{\$\#}$
		P2	164.2 ± 63.7	211.0 ± 72.2$^{\$\#}$	1–0 ± 0.3	1.3 ± 0.3$^{\$\#}$	37.2 ± 39.0	59.0 ± 55.0$^{\$\#}$
	DEC (n)	P1	138.8 ± 28.9	228.9 ± 57.3$^{\$♣\#}$	0.9 ± 0.2	1.4 ± 0.3$^{\$\#♣}$	27.4 ± 33.8	65.1 ± 57.1$^{\$\#♣}$
		P2	145.2 ± 50.6	197.7 ± 62.2$^{\$\#}$	0.9 ± 0.3	1.2 ± 0.3$^{\$\#}$	29.5 ± 33.7	51.2 ± 49.0$^{\$\#}$
4T-7D	TD (m)	P1	16,792.4 ± 3,222.9	20,917.4 ± 3,587.7$^{\$♣\#}$	1.3 ± 0.2	1.6 ± 0.3$^{\$\#♣}$	21,945.5 ± 7,023.7	33,323 ± 9,563.0$^{\$\#♣}$
		P2	17,165.1 ± 3,932.8	18,507.5 ± 3,514.4$^{\$\#}$	1.4 ± 0.3$^{\$}$	1.5 ± 0.7$^{\$\#}$	24,580.8 ± 9,468.9$^{\$}$	28,090.3 ± 7,480.2$^{\$\#}$
	PL (au)	P1	1,730.3 ± 383.5	2203.1 ± 416.6$^{\$\#♣}$	1.3 ± 0.2	1.6 ± 0.3$^{\$♣\#}$	2,286.8 ± 816.7	3,550.8 ± 1,053.3$^{\$\#♣}$
		P2	1,823.7 ± 359.6	1995.5 ± 370.7$^{\$\#}$	1.4 ± 0.2$^{\$}$	1.5 ± 0.3$^{\$\#}$	2,595.4 ± 824.2$^{\$}$	3,091.5 ± 765.4$^{\$\#}$
	HSR (m)	P1	422.4 ± 263.7	627.6 ± 276.9$^{\$\#♣}$	0.7 ± 0.3	0.9 ± 0.4$^{\$♣}$	385.8 ± 433.6	678.9 ± 477.8$^{\$\#♣}$
		P2	523.7 ± 216.7$^{\$}$	454.9 ± 222.2	0.9 ± 0.3$^{\#♣}$	0.7 ± 0.3	528.6 ± 359$^{\$♣}$	382.2 ± 309.8
	VHSR (m)	P1	137.5 ± 111.4	215.3 ± 124.6$^{\$\#♣}$	0.6 ± 0.4$^{\#♣}$	0.8 ± 0.4$^{\$\#♣}$	19.4 ± 54.3	35.3 ± 61.2$^{\$♣}$
		P2	138.4 ± 92.9	138.3 ± 96.4	0.6 ± 0.3	0.5 ± 0.3	17.6 ± 40.0	15.2 ± 32.0
	SPR (m)	P1	32.4 ± 39.7	53.7 ± 49.8$^{\$\#♣}$	0.4 ± 0.4$^{♣}$	0.6 ± 0.5$^{\$\#♣}$	4.4 ± 18.3	8.1 ± 22.3$^{\$\#♣}$
		P2	19.5 ± 22.1	24.7 ± 27.1	0.3 ± 0.3	0.3 ± 0.3	1.9 ± 7.9	2.2 ± 7.5
	ACC (n)	P1	198.1 ± 55.5	274.9 ± 77.1$^{\$\#}$	1.1 ± 0.2	1.4 ± 0.3$^{\$\#}$	37.9 ± 39.0	66.8 ± 55.0$^{\$\#}$
		P2	227.4 ± 77.3$^{\$}$	275.4 ± 85.7$^{\$\#}$	1.2 ± 0.3	1.4 ± 0.3$^{\$\#}$	48.7 ± 48.0	66.6 ± 56.2$^{\$}$
	DEC (n)	P1	180.8 ± 48.7	261.1 ± 70.5$^{\$\#}$	1.0 ± 0.2	1.3 ± 0.3$^{\$\#}$	32.8 ± 35.3	61.3 ± 51.1$^{\$\#}$
		P2	207.6 ± 47.6$^{\$}$	261.1 ± 68.0$^{\$\#}$	1.1 ± 0.2	1.3 ± 0.3$^{\$\#}$	40.1 ± 39.6	59.2 ± 50.3$^{\$\#}$

Note: 3T-6D is three training sessions in six days between matches; 4T-7D is four training sessions in seven days between matches; periodization type 1 (P1, post-match training session was conducted on MD+1); periodization type 2 (P2, post-match training session was conducted on MD+2), starter player and non-starter player. TD= Total distance; HSR= total distance >21 km·h^{-1}; VHSR = total distance >24 km·h^{-1}; SPR = total distance >27 km·h^{-1}; Player Load = PL; Acceleration = ACC and Deceleration = DEC. $ is >Starter/P1, # is > Starter/P2, ♦ is >Non-starter/P1 and ♣ is > Non-starter/P2.
Effect size: □<0.2, trivial; □<0.6, small; □<0.2, trivial; □<1.2, moderate; □<2.0, large; and□≥ 2.0, very large.

to those using strategy P2. However, in 3T-6D, starters using strategy P1 had higher values than those using P2. In MD-1, starters using strategy P1 in 4T-7D had higher TD, VHSR, and ACC compared to those using strategy P2 in 3T-6D. Moreover, starters using strategy P1 in 4T-7D had higher VHSR than in 3T-6D with the same strategy.

4. Discussion

The main goal of this study was to compare the accumulated TL and its distribution (TM and TS) in competitive micro-cycles. We differentiated between two groups of players (starters and non-starters) using two post-match planning strategies (P1 and P2). The analysis considered 6-d and 7-d turnarounds between matches (3T-6D and 4T-7D, respectively). The results revealed that: 1) Non-starters trained more than the starters; 2) the week length (3T-6D and 4T-7D) influenced external demands in both periodization types and player status; 3) P1 allowed non-starters to accumulate higher TL compensating for the lack of competition stimulus, maintaining similar TM and TS to P2; 4) P2 led starters to accumulate higher TL with elevated TM and TS levels.

As expected, for all players (e.g., starters and non-starters), the 4T-7D accumulated higher TL values than the 3T-6D. In contrast to previous studies (Anderson et al., 2016; Stevens et al., 2017) non-starters accumulated more TL than starters in the present investigation. This difference is because the competition TL was not included in the present work. The decision not to include the competition load was made to focus exclusively on the pre-competition TL, making it possible to assess the degree of acute fatigue or freshness with which the players arrived at the next match (Guridi et al., 2021) but for this study, using the estimation of the indicators of monotony and strain.

The non-starters accumulated a higher TL than the starters in both micro-cycle lengths (3T-6D and 4T-7D) and types of periodization (P1 and P2). This trend was consistent, except for the locomotor variables (HSR, VHSR, and SPR) in 4T-7D. P2, where no significant differences were observed between players (Figures 1 and 2). In line with Teixeira et al (Teixeira et al., 2022) monitoring only training session load, also described that non-starter accumulated higher high-intensity activity, especially in HSR, SPR and DEC metrics. The non-starters in P1 accumulated a higher TL than in P2 in all the metrics studied. In contrast, the starters accumulated higher TL in P2 than in P1. These results suggest that the type of periodization proposed (P1 or P2) could affect differently the accumulated TL by the player depending on his status.

In relation to the weekly distribution of TL (e.g., TM and TS), differences were found between players of different status, depending on the type of micro-cycle periodization and the number of days between matches (see Table 2). In line with Garcia et al. (2022) where it was observed that players with <60 min of participation in the competition showed higher values of TM, in the present study (from an overview) it was also observed that the non-starters presented for both indicators,

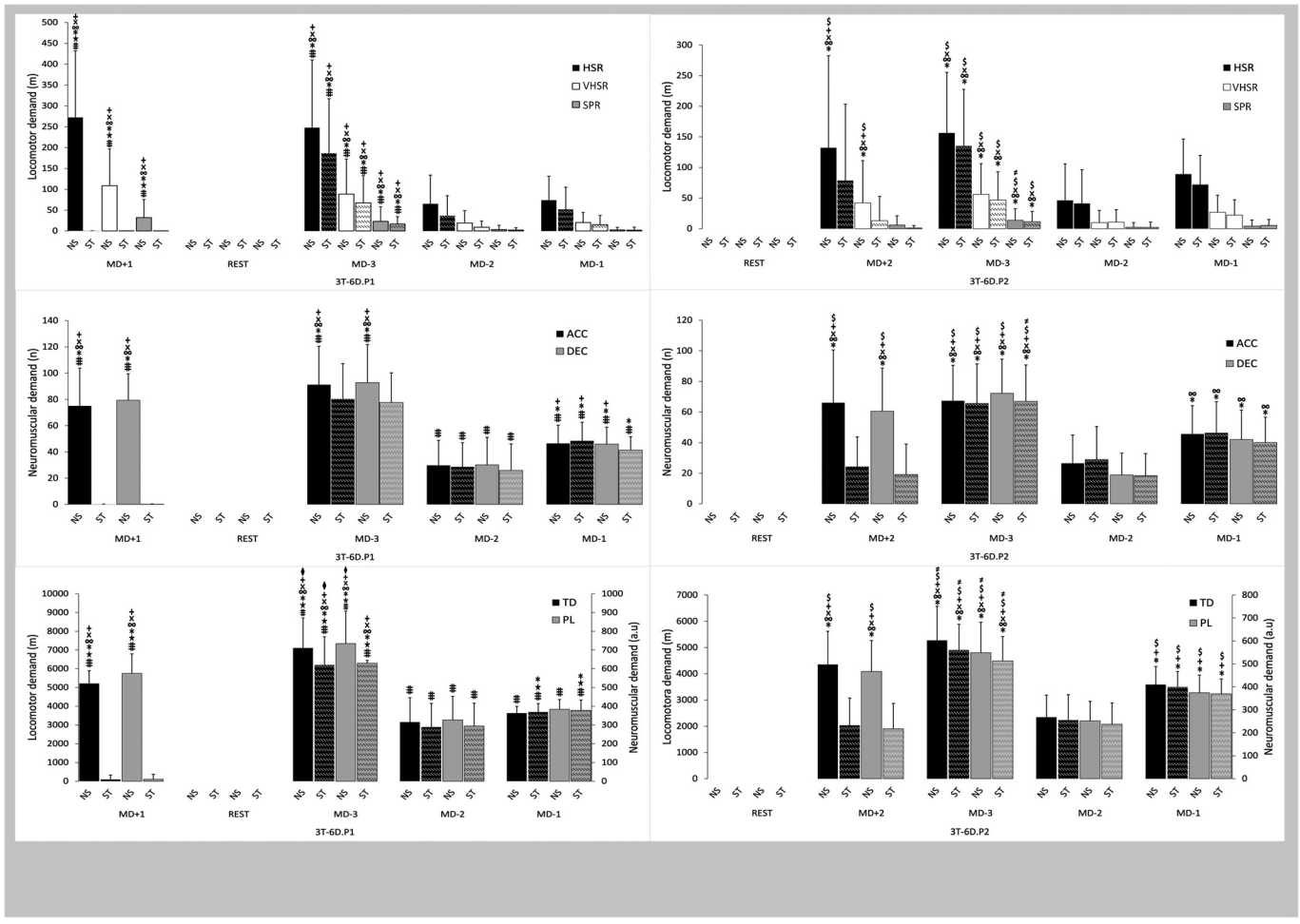

Figure 1. Descriptive statistics of locomotor and neuromuscular weekly demand (mean and standard deviation) of starting (ST) and non-starting (NS) players between two different types of micro-cycles (3T-6D) periodization (P1 and P2).Note: a) 3T-6DP1= post-match training in D+1 and rest in D+2, and 3 pre-match training sessions; b) 3T-6DP2= rest in D+1 and post-match training in D+2, and 3 pre-match training sessions. PL = Player Load (arbitrary unit, au), TD= Total Distance (m); HSR = total distance (m) >21 km·h^{-1}; VHSR = total distance (m) >24 km·h^{-1}; SPR = total distance (m) >27 km·h^{-1}, ACC= Acceleration (n) and DEC= Deceleration (n). Significant differences (*p <0.05*) are showed as: **♯** is >ST/MD+1; **$** is >ST/MD+2; **∂** is >ST/MD-4; * is >ST/MD-3; * is >ST/MD-2; **×** is >ST/MD-1; **Σ** is >NS/MD+1; **≠** is >NS/MD+2; **&** is >NS/MD-4; **∞** is >NS/MD-2; **✚** is >NS/MD-1.

TM and TS, significantly higher values than the starters. Particularly, in P1, the non-starters presented higher values of TM and TS compared to the starters. For the starters, the P2 strategy implied an increase in TM and TS values for the micro-cycle. That is, while the P1 increased the TM and TS for the non-starters, in P2 these differences were reduced with respect to the starters. This aspect is important because a similar continued demand of TL between training sessions could favour the appearance of residual fatigue, increasing the likelihood of injury (Castillo et al., 2019) especially in the players who could play in the next match. In the weekly periodization, it is important to consider the player's status, the location of the post-match session and the length of the micro-cycle to manage the distribution of TL, with which to minimize the cumulative effects of fatigue (Kelly & Coutts, 2007) and reduce, as much as possible, physiological interference (Buchheit et al., 2018). This could help not to compromise the physical performance of players in the competition. As expected, in both players status (starters and non-starters), 4T-7D elicited higher TM and TS values than 3T-6D.

The differences in TL, TM, and TS between starters and non-starters during the micro-cycle can be explained by the values observed in the training sessions. First, it is worth noting that the conditional demands of the training sessions (TD, HSR, VHSR, ACC, and DEC) align with reference ranges described in the literature for professional teams (Oliveira et al., 2021) 2,348–9,634 versus 2,143–9,540 (m in TD), 366–475 versus 7–541 (m, HSR vs TD18), 111–155 versus 1–190 (m in VHSR), 184–197 versus 9–195 (n of ACC) and 77–199 versus 10–157 (n of DEC), respectively. On the other hand, the variance in accumulated TL between starters and non-starters is mainly due to compensatory sessions for non-starters and recovery sessions for starters, along with a rest day. During the two days after the match, it is common for professionals in team sports to have this compensation/recovery phase (Cross et al., 2019). For instance, in MD+1 of 3T-6D.P1, non-starters had higher demand in PL, TD, HSR, VHSR, and SPR compared to starters, as it was the session designed to compensate for the lack of match stimulation. Managing TL on the match day

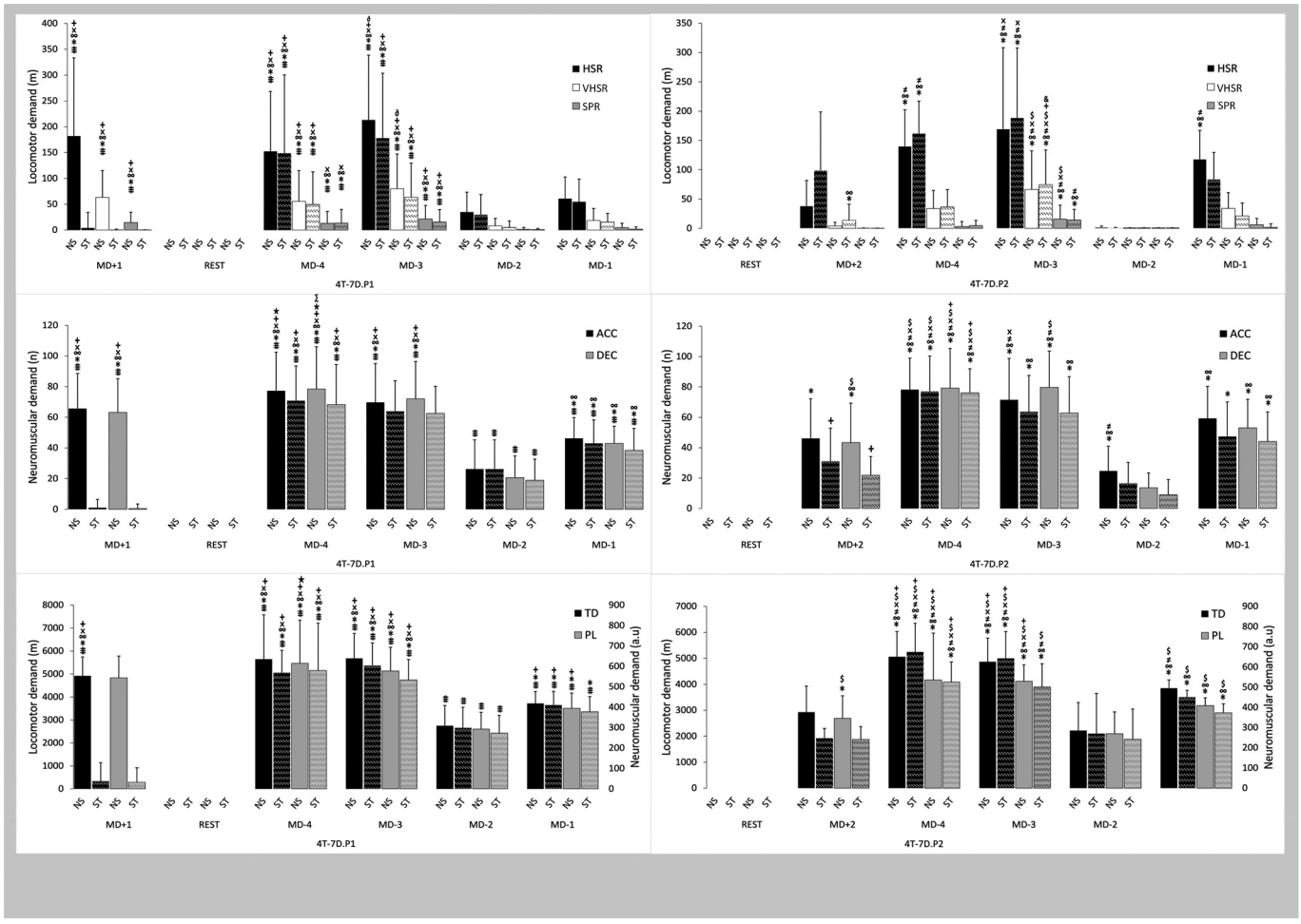

Figure 2. Descriptive statistics of locomotor and neuromuscular weekly demand (mean and standard deviation) of starting (ST) and non-starting (NS) players between two different types of micro-cycles (4T-7D) periodization (P1 and P2). a) 4T-7DP1= post-match training in D+1 and rest in D+2, and 4 pre-match training sessions; and D) 4T-7DP2= rest in D+1 and training in D+2, and 4 pre-match training sessions. PL = Player Load (arbitrary unit, au), TD= Total Distance (m); HSR = total distance (m) >21 km·h^{-1}; VHSR = total distance (m) >24 km·h^{-1}; SPR = total distance (m) >27 km·h^{-1}, ACC= Acceleration (n) and DEC= Deceleration (n). Significant differences (*p <0.05*) are showed as: **♯** is >ST/MD+1; **$** is >ST/MD+2; **∂** is >ST/MD-4; ***** is >ST/MD-3; ***** is >ST/MD-2; **×** is >ST/MD-1; **Σ** is >NS/MD+1; **≠** is >NS/MD+2; **&** is >NS/MD-4; **∞** is >NS/MD-2; **✚** is >NS/MD-1.

and the following two days is double challenging for the coaching staff to ensure adequate stimulus for non-starters while respecting starters recovery window (Casamichana et al., 2022).

In the MD-3 and MD-2 sessions, differences were also observed for the starters in 3T-6D.P1 with respect to the non-starters in both micro-cycle lengths and periodization strategies, with higher values for the variables PL, TD, HSR, ACC and DEC. The strategy chosen by the coaching staff seems to indicate that despite being a short week (three training sessions prior to the next match) and given the low probability of the non-starters becoming starters in the next match, they opted to supplement the conditional demand of this group of players in the central sessions of the micro-cycle, as has been described in other teams and leagues (Oliveira et al., 2021). These results highlight the importance of paying attention to the management of loads according to the player's status, in order to avoid substantial changes when following micro-cycles configured with different amounts of days between games (e.g., 6 or 7 days shown in the present study). According to previous studies (Cross et al., 2016; Ehrmann et al., 2016; Gabbett et al., 2016) there is evidence to suggest that an increase in the load placed on players may be linked to a higher risk of injury.

Although the present study provides valuable insights into TL accumulated by starters and non-starters in two micro-cycle lengths (3T-6D and 4T-7D) with two periodization strategies (P1 and P2) for the same team, it has some limitations. Generalizing the results is restricted due to the observational nature of the study and the possibility of different outcomes based on other training philosophies, even though the team analysed reported similar values to those in the literature (Oliveira et al., 2021). Further case studies are needed to solidify knowledge on training periodization strategies in professional soccer. Secondly, the study is limited to only two types of micro-cycles with 6- or 7-day rotations, overlooking variations in micro-cycle structures due to match congestion and team travel. Internal player response, including heart rate-derived measures, perceived effort, and recovery or wellness data (Guridi et al., 2021) was not considered, which could have provided valuable insights into individual player acute responses (Suarez-Arrones et al., 2015) and readiness to compete. External factors like travel, season timing, and match schedules may also influence internal

response. Therefore, more research is required to confirm or refute the study's findings on TL accumulation and distribution during 6-day and 7-day micro-cycles in football.

5. Practical applications

As a team staff, it is crucial to anticipate the training load (TL), match load (TM), and stress load (TS) that the periodization strategy of the competitive week may entail. This helps to choose the appropriate training programme that balances stimulus and recovery without neglecting the players' fitness in the long term or compromising their performance in the short term. Based on our research, we recommend the following practical applications: 1) If the conditions allow it, and the compensatory session for the non-starters players could not be carried out on the match-day, we suggest training on MD+1 (the day after the match) for non-starters. This helps to bring the absence of competitive conditional stimulation closer in time and facilitates the recovery process in starters. 2) Training on MD+2 (two days after the match) is an opportunity for starters to accumulate more load, especially in the first few competitive micro-cycles, where players are still carrying the load accumulated during the beginning of the competitive season. 3) The distribution of training content becomes essential when moving from a 4-day training rotation to a 3-day rotation within the micro-cycle or vice versa. We suggest redistributing the programmed content over three or two days to ensure training effectiveness, prevent detraining, and avoid peaks in the load to optimize training adaptations.

6. Conclusions

The study findings reveal the differences in training load (TL) between starters and non-starters, based on the type of periodization and the number of training days available between matches. The study found that non-starters accumulated higher TL than starters, regardless of the training process; this was true for both types of periodization, and the length of the week. Additionally, longer micro-cycles led to an increase in TL for both groups of players. In P1, non-starters accumulated higher TL, with similar values of TM and TS in both micro-cycle lengths, while in P2, starters accumulated higher TL, accompanied by higher values of TM and TS. Finally, the study found that the P1 plan generated higher conditional demands for non-starters, involving a redistribution of conditional content to optimize performance in the micro-cycle with fewer sessions (3T-6D).

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